

RISK & RETURN IN INVESTING

Financial Mathematics • CA-III • Unit V and Unit VI

Name: _____

PRN / Roll No: _____

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UNIT V: BASICS OF INVESTMENT

1. Investment Return

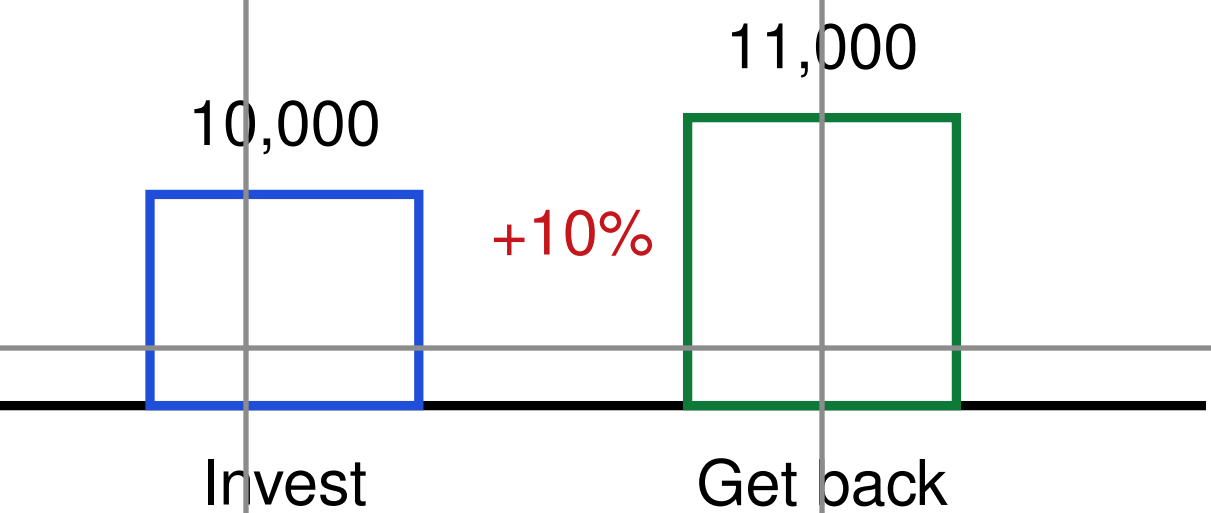
The gain (income + price rise) on an investment, as a % of the money invested.

$$\text{Return} = \frac{\text{Gain}}{\text{Invested}} \times 100$$

Eg: Invest Rs 10,000, get back Rs 11,000.

$$\text{Return} = 1,000 \div 10,000 = 10\%$$

Real life: returns on shares and mutual funds.



2. Compounding Frequency

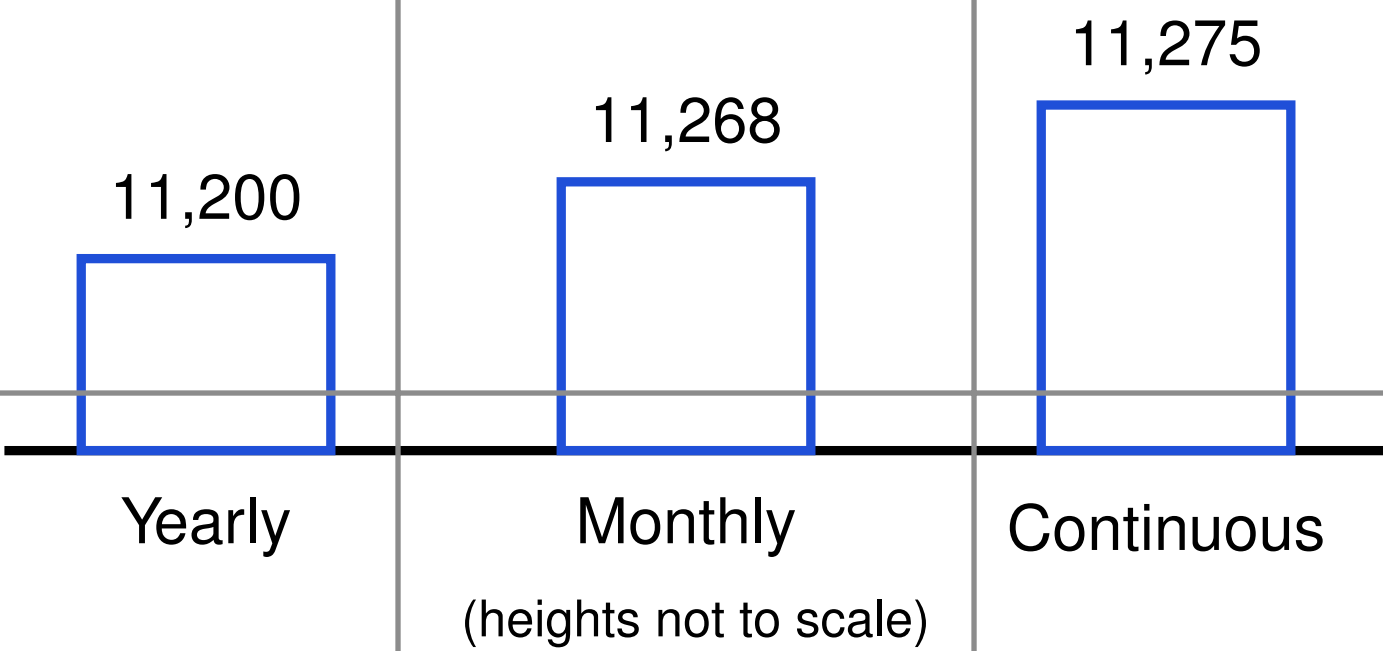
At the same yearly rate, adding interest more often gives a little more money.

$$A = P \left(1 + \frac{r}{m} \right)^{mt}$$

$$\text{Continuous: } A = Pe^{rt}$$

Eg: Rs 10,000 at 12% for 1 year

Real life: bank FDs add interest quarterly.



3. Uneven Cash Flows

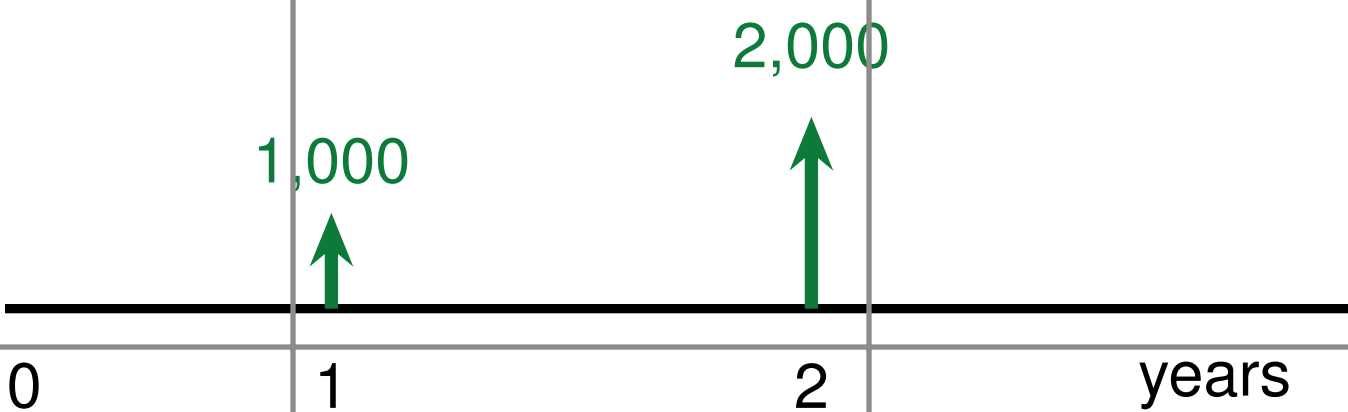
Money received differs each year. Discount each to today and add.

$$PV = \sum \frac{CF_t}{(1+i)^t}$$

Eg: Rs 1,000 (year 1) and Rs 2,000 (year 2), $i = 10\%$

$$PV = 909 + 1,653 = \text{Rs } 2,562$$

Real life: rent, dividends or project profits.



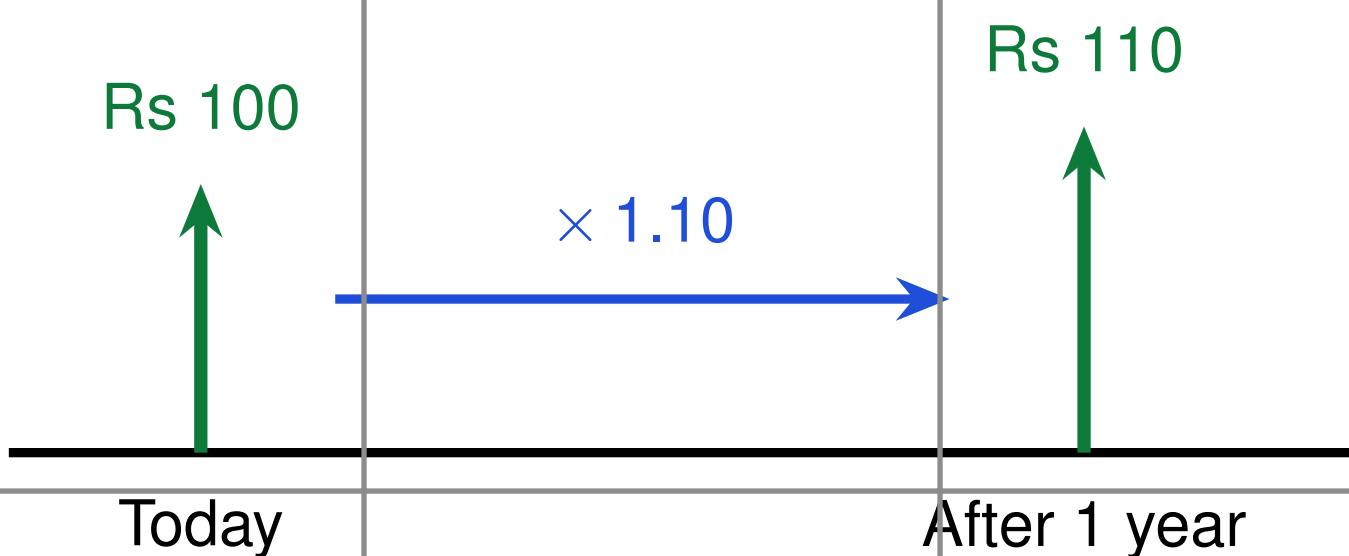
4. Economic Equivalence

At a given interest rate, different sums at different times can have the same value.

$$F = P(1+i)^n \quad P = \frac{F}{(1+i)^n}$$

Eg: at 10%, Rs 100 now = Rs 110 in 1 yr

Real life: lump sum today vs instalments.



UNIT VI: RISK AND UNCERTAINTY

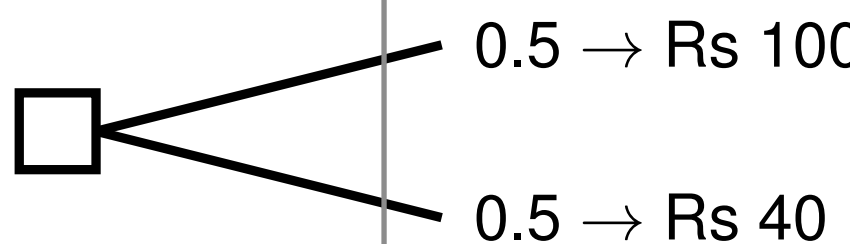
1. Decision under Risk

Chances are known. Choose the best average (EMV).

$$\text{EMV} = \sum (\text{chance} \times \text{payoff})$$

Eg: $0.5 \times 100 + 0.5 \times 40 = \text{Rs } 70$

Real life: choosing which project to fund.



2. Decision under Uncertainty

Chances are unknown. Use a rule:

- **Maximax:** best of the best → **Stock**
- **Maximin:** best of the worst → **Bond**
- **Min-max regret:** least regret → **Bond**

Payoff	Boom	Slump
Stock	80	-20
Bond	40	25

Real life: launching a product in a new market.

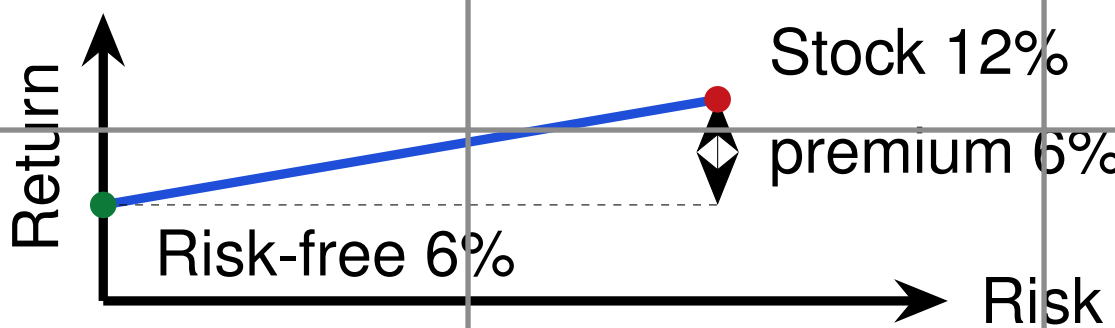
3. Risk Premium

Extra return for taking risk.

$$\text{Premium} = \text{Expected} - \text{Risk-free rate}$$

Eg: $12\% - 6\% = 6\%$

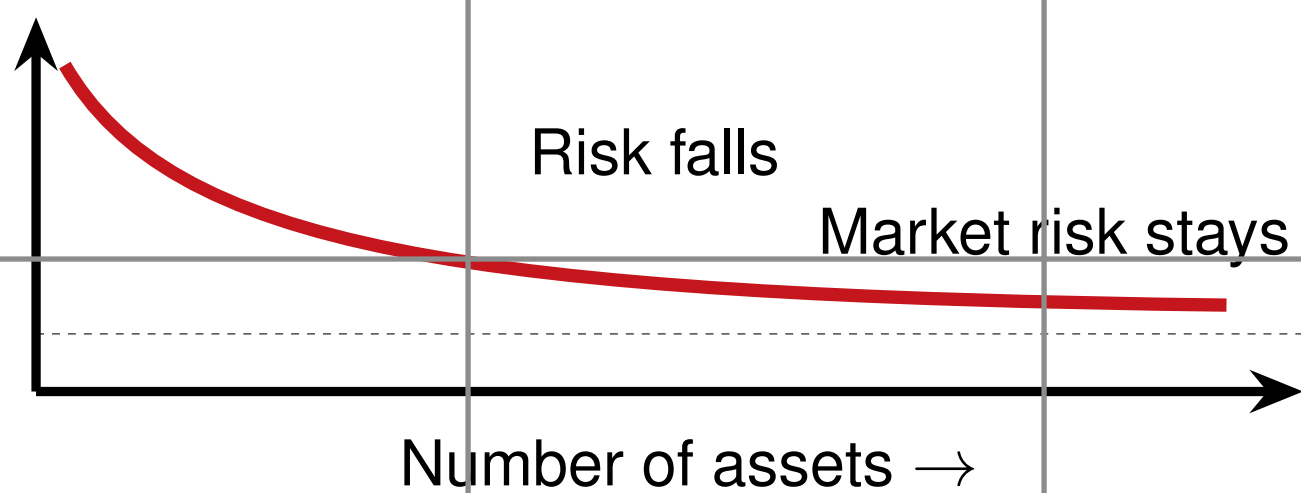
Real life: shares must beat the FD rate.



4. Portfolio Diversification

Spread money over assets that do not move together. Market risk stays.

Real life: mutual funds hold many shares.



5. Life Insurance

You pay a premium. The insurer pays the sum assured if you die within the term.

$$\text{Premium} \approx \text{Sum assured} \times P(\text{death})$$

Eg (1 yr): $\text{Rs } 1,00,000 \times 0.01 = \text{Rs } 1,000$

Real life: term life plans.



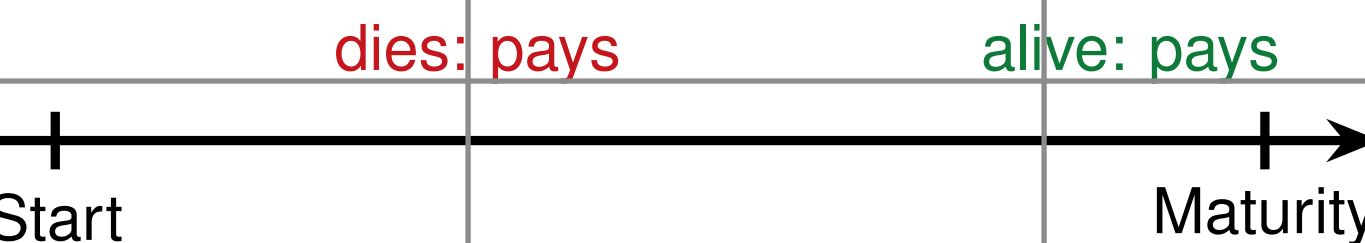
6. Endowment

Pays the sum assured if you **die in the term** OR **survive to the end**.

$$\text{Endowment} = \text{Term} + \text{Pure endowment}$$

Pure endowment = paid only if you survive.

Real life: endowment plans (cover + savings).



Investors demand a higher expected return for taking higher risk